



Enhancing Medical NER from clinical notes using LLMs

Clinical notes are unstructured with narrative text and many abbreviations. Doctors spend nearly 50% of their time on electronic health records (EHRs).

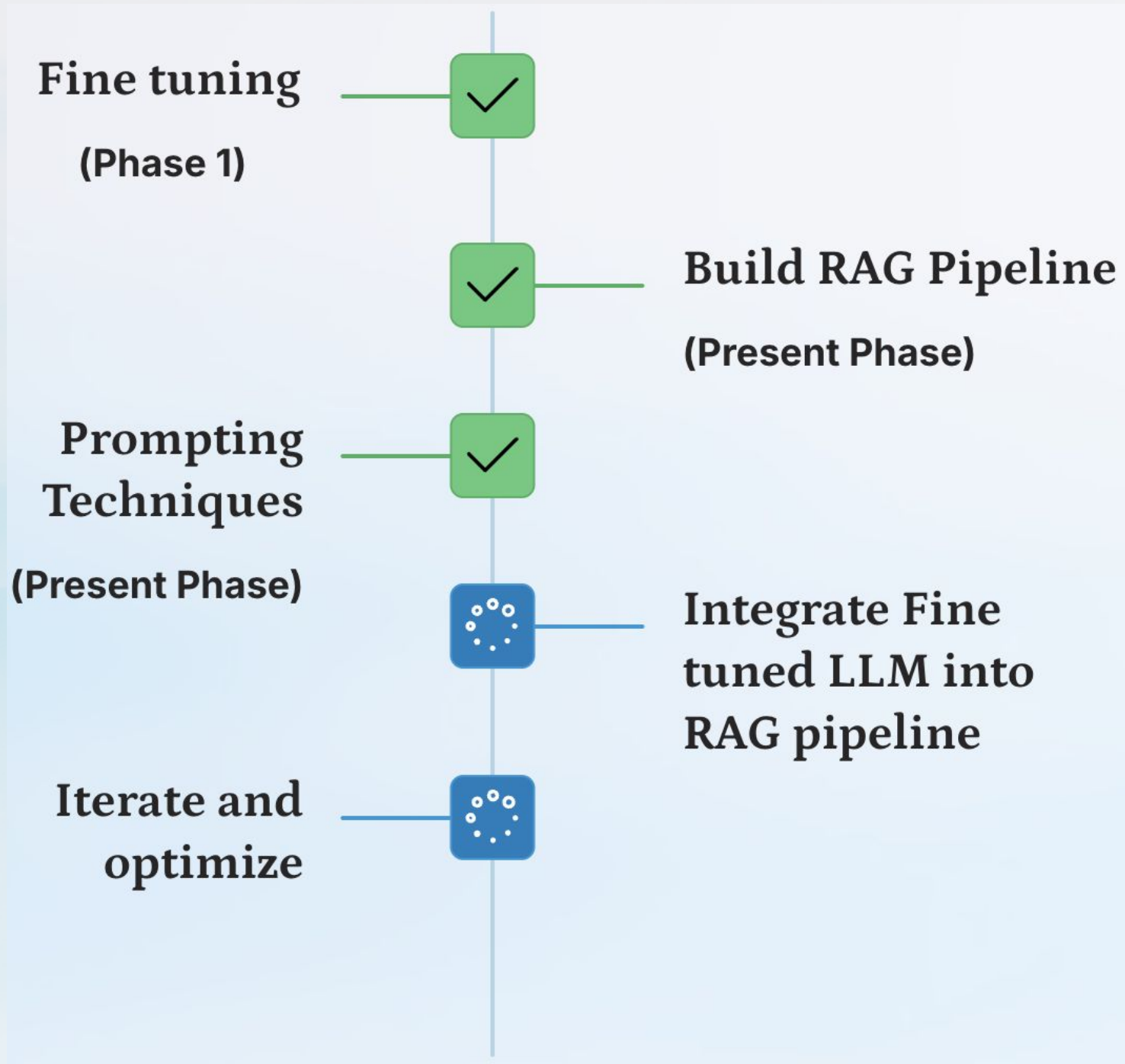
Our goal is to extract accurate, essential information like medications and conditions efficiently from these notes.

Team members

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Progress



What we have already completed

Fine tuned BERT for NER - SFT & PEFT

Medical Entity Extraction

AGE, GENDER, SYMPTOMS, LAB TESTS, LAB RESULTS, DIAGNOSIS, TREATMENT, MEDICATIONS

Weighted F1 score - 0.70

Entity mapping

**Symptoms: "SOB", "Asthma"
Diagnosis: "C3 Dermatome tear"
Treatment: "BRAAF", "ASA"**

Accuracy: 82%

Gaps:

Diagnosis: C3

What is C3?

Treatment: BRAAF

BRAAF for?

Diagnosis: COPD

I have never heard of COPD :(

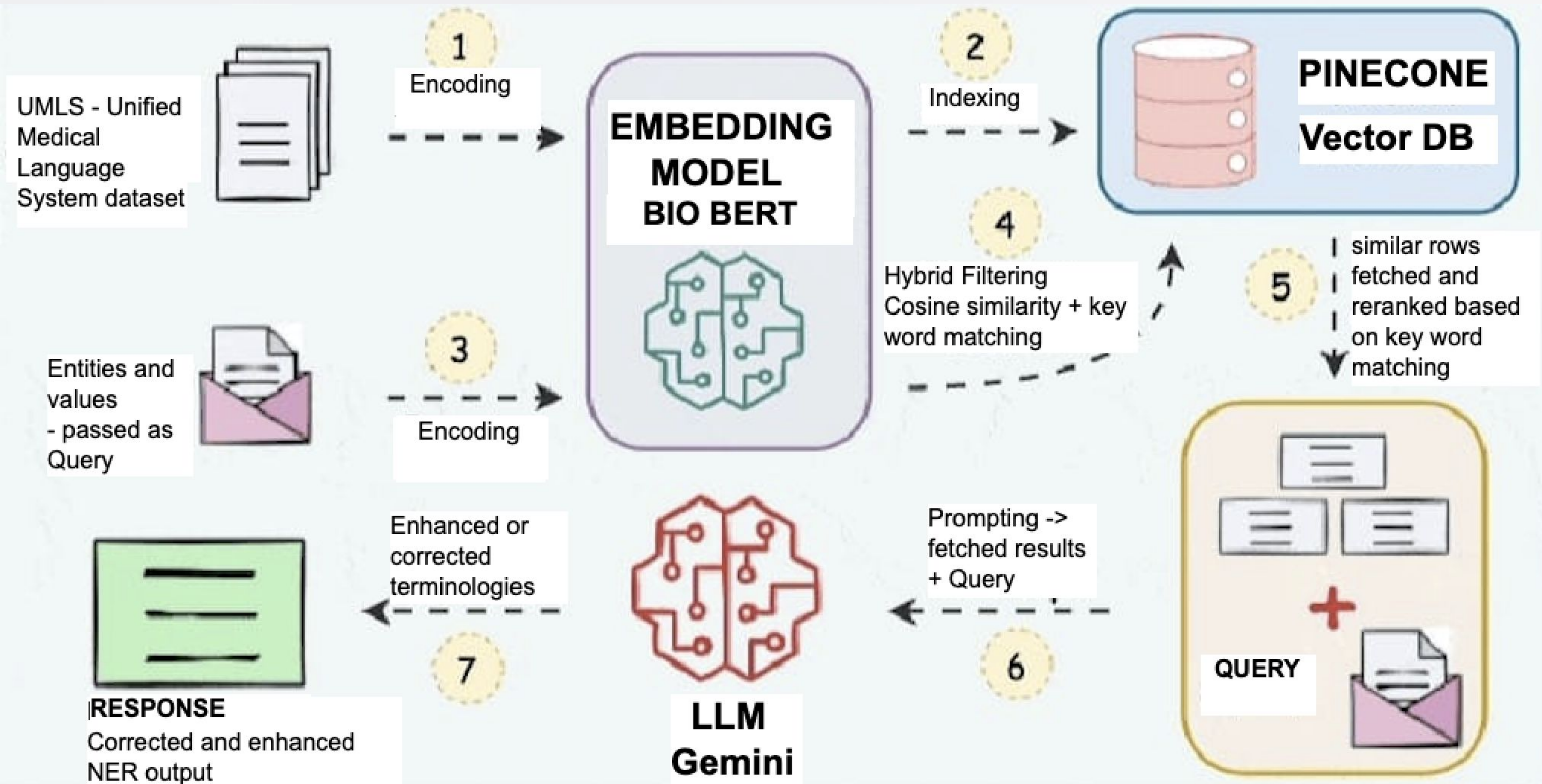
Dataset

ULMS: [Unified Medical Language System](#)

ENTITY	DEFINITION	ALIASES	NAME
string · lengths	string · lengths	string · lengths	string · lengths
			
8100%	3↔85028.3%	2↔1.37k77.6%	2↔201100%
C0732279	A drug combination of aspirin and dipyridamole that functions as a PLATELET AGGREGATION INHIBITOR, used to prevent THROMBOSIS and STROKE in TRANSIENT ISCHEMIC ATTACK patients.	Aspirin- and dipyridamole-containing product Combination, Aspirin-Dipyridamole Drug Product containing aspirin and dipyridamole (medicinal product) Drug Combination, Aspirin-Dipyridamole Aspirin, Dipyridamole Drug Combination Aspirin/Dipyridamole ASPIRIN/DIPYRIDAMOLE Aspirin-Dipyridamole Drug Combination Aspirin Dipyridamole Drug Combination	Aspirin / Dipyridamole

470k rows, Annotated

RAG PIPELINE



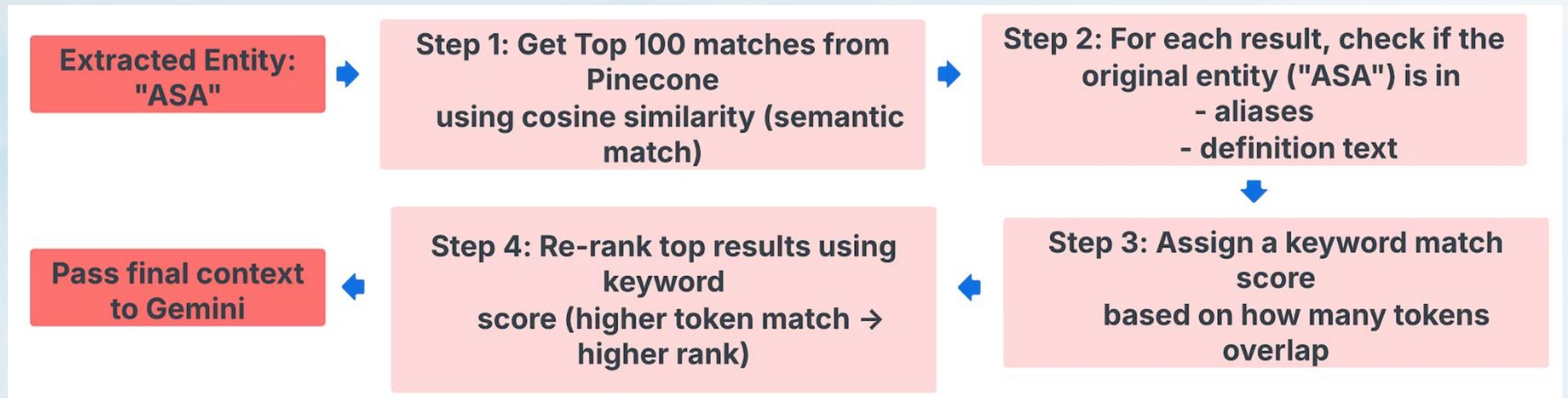
RAG

Vector database: Pinecone

Embeddings: BioBERT

Fetching top K results - Re-Ranking

- Cosine Similarity + **Keyword matching**



Prompting Techniques used to enhance the NER Output

3-shot prompting

```
prompt = f"""
You are an assistant that corrects clinical NER outputs using external medical context.

Use the examples below to guide your output.

---

Example 1:
Entity: ASA (Diagnosis)
Context: ASA is used to refer to Aspirin in cardiovascular notes.
→ Corrected: Aspirin
→ Reason: ASA is a common abbreviation for Aspirin in medicine.

---

Example 2:
Entity: SOB (Symptoms)
Context: SOB refers to shortness of breath.
→ Corrected: Shortness of breath
→ Reason: SOB is a standard abbreviation.

---

Example 3:
Entity: LFT (Diagnosis)
Context: LFT refers to liver function tests such as ALT, AST, and bilirubin.
→ Corrected: Liver Function Test Abnormalities
→ Reason: LFT is an abbreviation commonly used for liver enzyme evaluations.

---

Now process the following:

Entities:
{entity_string}

Context:
{full_context}

Return each entity's:
- Corrected version
- Reason for correction
- Output as structured JSON grouped by original labels.
"""
```

Retrieve-Task-Example prompting

```
prompt = f"""
You are a highly reliable clinical NLP assistant with expertise in Named Entity Recognition (NER) enhancement using external knowledge.

Your task is to receive potentially noisy or unclear biomedical entities extracted from clinical notes, and correct or clarify them using the provided medical knowledge context.

--- Example ---

Input:
Entity: ASA (Diagnosis)
Context: ASA is widely used to refer to Aspirin in cardiovascular notes.

Output:
Corrected: Aspirin
Reason: ASA is an abbreviation of Aspirin.
|
--- End of Example ---

Now enhance the following entities using the context provided:

Entities:
{entity_string}

Context:
{full_context}

Please output:
- JSON containing enhanced entities grouped under their original labels
```


Instructions embedded with Chain-of-thought prompting

Instructions:

1: Understand the context

2: Enhance entities that are unclear.

3: extract the abbreviations.

For instruction 3, lets think for each entity:

- what the abbreviation or misspelt term could stand for.

- how the context helps you disambiguate it.

- Will the contextual meaning be retained if you replace the terms with standard terminology?

5. Replace the abbrevation with fullform without changing the contextual meaning

6: Normalize drugs, symptoms, or diseases using the context.

For example: chemo becomes chemo therapy for treatments, chemo becomes chemo drugs for medication

7: Keep structure by returning enhanced entities grouped under original labels.

Return result like:

```
{{  
  "Symptoms": ["Shortness of breath", "Fever"],  
  "Diagnosis": ["Aspirin"]  
}}
```

Instructions embedded with few shot prompting

```
prompt = prompt = f"""
You are a clinical AI assistant.

You are given an NER output from a medical note with categories and entities.
Some may be abbreviated, misspelled, or unclear.

Entities:
{entity_string}

Knowledge Context:
{full_context}

Instructions:
1: Understand the context
2: Enhance entities that are unclear.
3: extract the abbreviations and its fullform. Replace the abbreviation with fullform
   For example:
   ASA is Aspirin, SOB is shortness of breath, BP is blood pressure
4: Normalize drugs, symptoms, or diseases using the context.
   For example:
   chemo becomes chemo therapy for treatments, chemo becomes chemo drugs for medication, Chemo becomed chemo test for Lab tests
5: Keep structure by returning enhanced entities grouped under original labels.

Return result like:
{{
  "Symptoms": ["Shortness of breath", "Fever"],
  "Diagnosis": ["Aspirin"]
}}
"""
```

LLM Output Without RAG (No External Knowledge)

```
test_ner = {  
    "Symptoms": ["SOB", "weakness in wrist flexion"],  
    "Diagnosis": ["Dermatome of C3"]  
}  
  
print(enhance_ner_output(test_ner))
```

```
```json  
{
 "Symptoms": ["Shortness of breath", "Weakness in wrist flexion"],
 "Diagnosis": ["Dermatome of C3"]
}
```
```

WITH RAG

```
test_ner = {  
    "Symptoms": ["SOB", "weakness in wrist flexion"],  
    "Diagnosis": ["Dermatome of C3"]  
}  
  
print(enhance_ner_output(test_ner))
```

```
```json  
{
 "Symptoms": ["Shortness of breath", "Wrist flexion weakness"],
 "Diagnosis": ["Dermatome of the third cervical nerve"]
}
```
```


LLM Output Without RAG (No External Knowledge)

```
✓ 0s  test_ner = {  
    "Symptoms": ["Common cold", "weakness in wrist flexion"],  
    "Treatment": ["BRAf"]}  
  
print(enhance_ner_output(test_ner))  
  
 ```json  
{  
  "Symptoms": ["Common cold", "Weakness in wrist flexion"],  
  "Treatment": ["BRAf"]  
}
```

WITH RAG

```
✓ 1s  test_ner = {  
    "Symptoms": ["Common cold", "weakness in wrist flexion"],  
    "Treatment": ["BRAf"]}  
  
print(enhance_ner_output(test_ner))  
  
 ```json  
{  
  "Symptoms": ["Common cold", "Wrist flexion weakness"],  
  "Treatment": ["BRAf targeted therapy"]  
}
```

FUTURE WORK



Refine Fine-Tuning

Continue training and hyperparameter tuning to improve **F1-score** and reduce false positives/negatives in entity extraction.



Integrate with RAG Pipeline

Seamlessly connect the fine-tuned BERT output into the RAG-based entity enhancement system.



Validate Results

Test the integrated pipeline on clinical notes and evaluate improvements in **accuracy** and **completeness**



Iterate & Optimize

Use feedback and evaluation metrics to continuously refine the NER model and RAG retrieval strategies.



THANK YOU